Optimal Solar Panel Battery Management at the Household Scale

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Abstract. As solar panels and home batteries become more widely available, the question of how to best manage energy storage is becoming increasingly important. A household equipped with these systems must constantly decide: should it charge the battery or feed its excess production back into the grid? The challenge stems from uncertainty regarding consumption, solar production, and electricity prices. We propose a realistic simulation environment using stochastic differential equations coupled with a deep reinforcement learning algorithm to automatically learn an optimal management strategy.

Keywords: Deep Reinforcement Learning · Stochastic models · Battery control.

1 Introduction

The rapid deployment of residential photovoltaic systems has transformed households into small-scale energy producers. In the past, surplus electricity could be resold to EDF at a fixed regulated tariff, providing predictable revenues and reducing the need for complex optimisation strategies. Today, feed-in tariffs are less advantageous or more regulated, and households increasingly rely on self-consumption and battery storage to maximise the value of their production. At the same time, solar production remains variable and electricity prices can fluctuate due to time-of-use or dynamic market mechanisms. These factors make optimal battery management a non-trivial sequential decision problem.

This report is organized as follow : in **Section 1** the environment and models that it uses are detailed. The Deep Q Learning algorithm used is presented in **Section 2**. Experimental results are presented in **Section 3** and discussed in **Section 4**. Code can be accessed at https://github.com/ychv/Optimal-Solar-Panel-Battery-Management-at-the-Household-Scale

2 Environment and Models

2.1 Actions, observation spaces and rewards

The environment action space is composed of 3 actions : charging the battery, discharging it, or stay idle. The observation space is composed of the battery

current charge, the current time step, and a forecast (from current time to x steps ahead) of the house electric consumption, the house solar production, and electricity price. The agent is rewarded proportionally to the quantity of electricity it buys or sells, depending on the quantity needed or in excess (production - consumption). Consumption, production and price forecasts are updated in a first-in first-out fashion according to the models described in the following section. It is worth noting that the battery is charged and discharged in a single time step.

2.2 Models

Consumption To simulate household electricity consumption, we need a process that is both realistic and stochastic. A typical household exhibits a characteristic daily profile: low consumption at night, with peaks in the morning and evening. However, consumption varies from day to day due to behavioural randomness, which motivates modelling it as a deterministic mean profile subject to stochastic fluctuations.

We model consumption as a mean-reverting Ornstein-Uhlenbeck process:

$$dL(t) = \theta \left(\mu(t) - L(t)\right) dt + \sigma(t)\, dW(t) \tag{1}$$

where $L(t)$ is the instantaneous power consumption (W), $\mu(t)$ is the deterministic mean profile, $\theta > 0$ controls the speed of mean reversion (relaxation time $\tau = 1/\theta$), σ is the deterministic standart deviation profile and $W(t)$ is a standard Wiener process. This formulation ensures that $L(t)$ fluctuates around a realistic daily shape while remaining physically interpretable.

The mean profile $\mu(t)$ is estimated empirically from the UCI *Individual Household Electric Power Consumption* dataset [2], which contains over 2 million one-minute measurements from a single household in Sceaux, France (2006–2010). The empirical average over all days yields a 1440-point profile capturing the characteristic morning and evening peaks.

Solar Production Solar irradiance follows a Gaussian intraday profile centered at solar noon, active between sunrise and sunset:

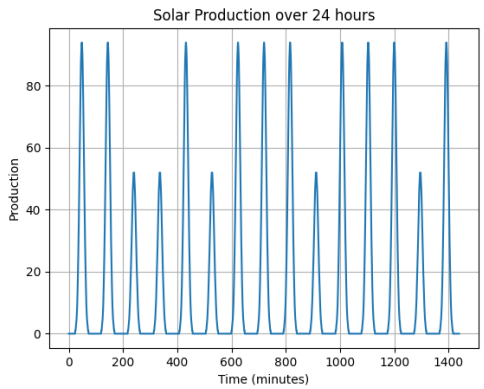
$$\mu_{\text{prod}}(t) = 420\ e_s\ \exp\left(-\frac{(t-720)^2}{2 \times 120^2}\right) \cdot \mathbf{1}_{[345,\,1095]}(t) \tag{2}$$

where t is expressed in minutes, $720\,\text{min} = 12\text{h}$, and $e_s \in \{0.1, 0.5, 0.9\}$ is a weather scaling factor representing overcast, cloudy, and sunny conditions respectively.

Day-to-day variability in weather is modelled by a first-order Markov chain over the three weather states, with transition matrix:

$$P = \begin{pmatrix} 0.6 & 0.3 & 0.1 \\ 0.3 & 0.4 & 0.3 \\ 0.1 & 0.3 & 0.6 \end{pmatrix} \tag{3}$$

The state is updated once per day, introducing realistic multi-day weather persistence. A sample of the resulting production profile over 15 days is shown in Figure 2.2.



Electricity Price The electricity price follows a deterministic time-of-use tariff inspired by the French peak/off-peak (heures pleines/heures creuses) structure:

$$p(t) = \begin{cases} 0.20\,€/\text{kWh} & \text{if } t \in [7\text{h}, 23\text{h}] \\ 0.10\,€/\text{kWh} & \text{otherwise} \end{cases} \tag{4}$$

3 Deep Q Learning

The control policy is optimized using Deep Q Learning [1], which consists in approximating the action-value function $q : (\mathcal{S}, \mathcal{A}) \mapsto R$ with a neural network. The model is trained on batches of action-state observations stored in a replay memory that is updated with each steps, To stabilize training, two networks are maintained: a policy network (with parameters θ) updated at every step, and a target network (with parameters θ') updated softly via

$$\theta' \leftarrow \tau\theta + (1-\tau)\theta' \tag{5}$$

The policy follows an ε-greedy strategy, decaying from full exploration ($\epsilon = 0.9$) to near-greedy exploitation ($\epsilon = 0.01$) over training. The network is optimized by minimizing the Huber loss between predicted q-values and the Bellman targets:

$$\mathcal{L} = E\left[\mathcal{H}_\delta(r + \gamma \max_{a'} q_{\theta'}(s', a') - q_\theta(s, a))\right] \tag{6}$$

4 Results

We first looked at what a random policy in this environment would yield. In figure 1 we can see the mean reward per episode (episodes being of fixed length mean

makes more sense here) with a battery of capacity of 100. Then we trained the DQL agent on different environment configurations (varying capacity, varying forecast depth) and different hyper-parameters configurations (varying ϵ decay and varying γ value, γ representing how short-sided (0) or far-sighted (1) the reward is considered by the agent). In all cases, all but the varying parameter are fixed to the same reference value of a battery capacity of 100, forecast of 10, $\gamma = 0.99$ and ϵ decay of 30 episodes, which corresponds to the blue curve.

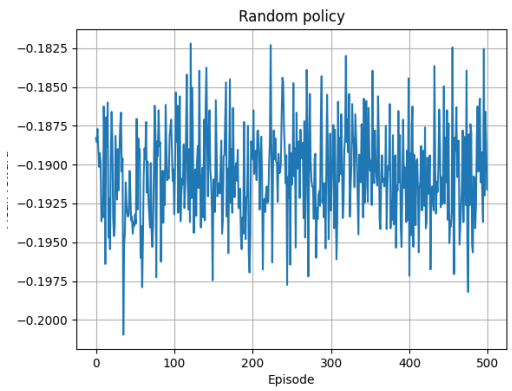
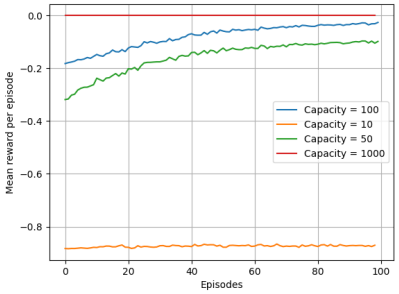
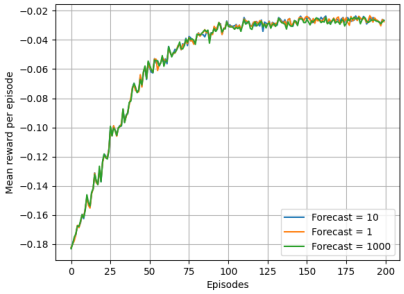


Fig. 1: Mean reward on 500 episodes with random policy

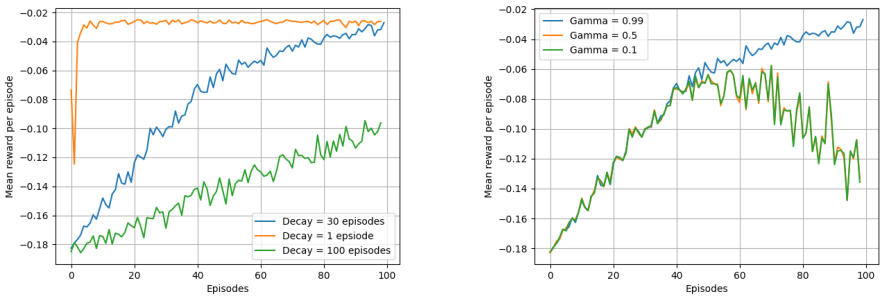


(a) Mean reward per episode against battery capacity



(b) Mean reward per episode against forecast depth

Fig. 2: Effect of varying environment variables on the agent



(a) Mean reward per episode against ϵ decay

(b) Mean reward per episode against γ value

Fig. 3: Effect of varying agent variables on the agent

5 Discussion

We see in fig 2a that a battery too small does not make a difference on the environment so the agent can't learn to monitor it. On the contrary, a huge battery, because of the assumption of instant charge and discharge, makes things too easy for the agent and allows instantly for almost no penalty. Interestingly, the forecast depth (fig 2b) seems to have no impact on learning : this could indicate that the agent is learning $\mu(t)$ (mean profile of the consumption model), as this function does not vary during training. A more precise model with seasons and especially random mean variation for example could lead to more sensibility to the forecast parameter.

As for agent hyper-parameters, we see in fig 3a that even with very little exploration the agent converges to the same mean reward as with a longer decay, adding to the assumption of overfitting the μ discussed previously. Finally in fig 3b we see that a γ too small, and thus a too short sighted consideration of the reward, leads the mean reward to drop after a few episodes.

References

1. Paszke, A. and Towers, M. : Reinforcement Learning (DQN) Tutorial. PyTorch (2017, upd 2025)
2. UCI Machine Learning Repository : Individual Household Electric Power Consumption Dataset. (2012)